Sangola College, Sangola
Dept. of Computer Science & Applications
B.Sc (ECS)-III (Data Mining using Weka)
Assignment No-02

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1) Generate Association Rules using the Apriori Algorithm. To find frequent item sets for above transaction with a minimum support of 2 having confidence measure of 70% (i.e, 0.7). Solve numerically and results should be checked within WEKA.

1) How many rules are generated?
2) find all frequent item sets
3) What are the best rules found?

TID	ITEMS
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5
500	1,4,5
600	3,4

2) To generate association rules using FP Growth Algorithm. To find all frequent item sets in following dataset using FP-growth algorithm. Solve numerically and results should be checked within WEKA. Minimum support=2 and confidence =70%

I. How many rules are generated?
II. find all frequent item sets
III. What are the best rules found?

TID	ITEMS
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5
500	1,4,5
600	3,4

3) Construct Decision Tree and generate the classification rules for the given dataset. Solve numerically and results should be checked within WEKA.

I. Load dataset into WEKA and run J48 classification algorithm.
II. How many instances are pruned?
III. Study the classifier output such as confusion matrix and derive Accuracy, F-measure, TPrate, FP-rate, precision and recall values.
IV. Compute entropy values, kappa statistic.
V. Use training dataset and implement decision tree using cross validation fold.
VI. Apply cross-validation strategy with various fold levels and compare the accuracy results.
VII. Draw Decision Tree.

Cross-validation fold
Correctly Classified Instances
Incorrectly Classified Instances
Mean absolute error
Root mean squared error
Relative absolute error
Root relative squared error
Total Number of Instances

Training Dataset:

Roll_No	Age	Income	Student	Credit_Rating	Buys_Computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle_aged	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle_aged	Medium	Yes	Excellent	Yes
8	Youth	Low	No	Fair	No
9	Youth	Medium	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle_aged	Medium	No	Excellent	Yes
13	Middle_aged	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Use Test Data: implement this test data in Weka

Roll_No	Age	Income	Student	Credit_Rating	Buys_Computer
15	Youth	High	No	Fair	No
16	Youth	High	No	Excellent	No
17	Middle_aged	High	No	Fair	Yes

4) Demonstration of Classification Rule Process on **species.csv** dataset using Navie Bayes Algorithm

Color	Legs	Height	Smelly	Species
White	3	Short	Yes	M
Green	2	Tall	No	M
Green	3	Short	Yes	M
White	3	Short	Yes	M
Green	2	Short	No	H
White	2	Tall	No	H
White	2	Tall	No	H
White	2	Short	Yes	H

Solve numerically and results should be checked within WEKA. Using the above data, we have to identify the species of an entity with the following data.

X={Color=Green, Legs=2, Height=Tall, Smelly=No}

I. Use **species.csv** as training dataset
II. Study the classifier output such as confusion matrix and derive Accuracy, F-measure, TPrate, FP-rate, precision and recall values.
III. Classify a new 'X' instance (Test dataset) based on the Naive Bayes model?

5) You have a training dataset of houses with features Size, Age, and price category. You need to predict whether a price of houses will be High, Medium or Low based on these features. Solve this problem numerically using K-Nearest Neighbors (KNN) Algorithm and results should be checked within WEKA. (Number of neighbor K=3)

Size (sq ft)	Age (years)	Price Category
1500	5	Medium
2000	10	High
1200	15	Low
1800	8	Medium
2500	6	High
1000	20	Low
2200	9	High
1300	12	Low

I. Study the classifier output such as Accuracy, F-measure, TPrate, FP-rate, precision and recall values and confusion matrix
II. Given a new house with Size = 1600 sq ft and Age = 7 years, you would predict the price category and performing majority voting.
III. Take k=5 and compare that result with previous result(k=3)

6) Use following training dataset and implement logistics regression algorithm. Solve this problem numerically and Weka tool.

Height(CM)	**Weight(Kg)**	T Shirt Size
158	58	M
158	59	M
158	63	M
160	59	M
163	61	M
160	64	L
163	64	L
165	61	L
165	65	L
168	62	L
168	63	L
170	63	L

I. Study the classifier output such as Accuracy, F-measure, TPrate, FP-rate, precision and recall values and confusion matrix
II. How many correctly and correctly classified instances?
III. What are the values of coefficients and intercept?
IV. Given a new data with Height(CM)=161 and Weight(Kg)=61, To predict what is the size of T-shirt.

7) Use above training dataset and implement Support Vector Machine algorithm.

I. Study the classifier output such as Accuracy, F-measure, TPrate, FP-rate, precision and recall values and confusion matrix
II. Compare results of Logistics regression and SVM?